Number

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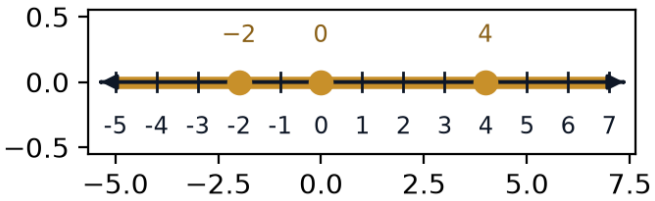
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In a class of 30 students, 15 students play football and 20 students play basketball. Every student plays at least one of the two sports. Using a Venn diagram, find the number of students who play both football and basketball.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A number line is shown below. What is the value of the number indicated by the arrow?



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The histogram shows the distribution of scores for a group of students in a mathematics test. The total number of students who scored between 0 and 40 is 32. The frequency density for the class interval $20 \le x < 40$ is 0.8. Calculate the number of students who scored between 20 and 40.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The diagram shows a number line. The distance between the points marked A and B is 15 units. The coordinate of A is -7. Find the possible coordinates of B.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A bag contains n marbles, of which 4 are green. Two marbles are taken at random without replacement. If the probability that both are green is $\frac{2}{15}$, find the total number of marbles n.

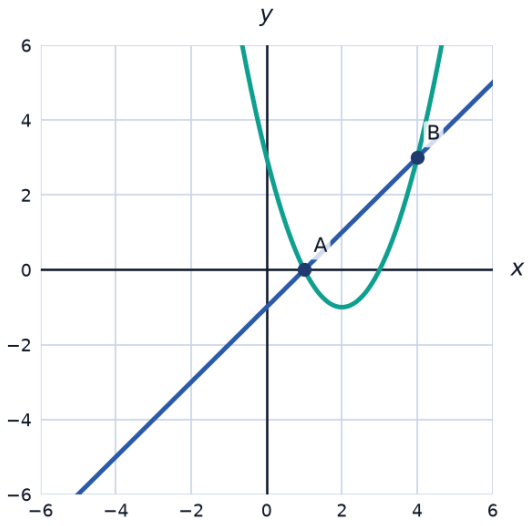
6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Set A has 12 elements and set B has 8 elements. If $A \cap Beq\varnothing$, what is the maximum possible number of elements in $A \cup B$?

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

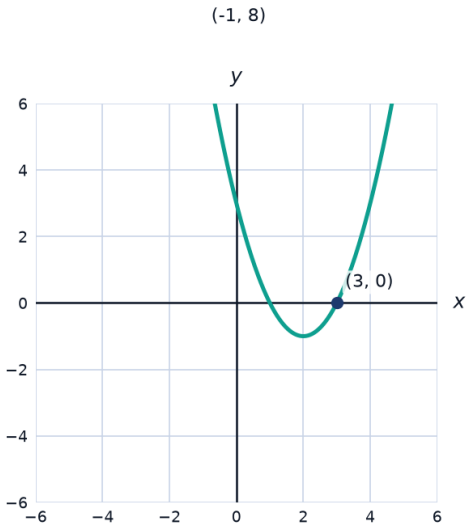
The graph of the quadratic function $y = x^2 - 4x + 3$ is shown. The graph of the linear function $y = x - 1$ is also shown. The graphs intersect at points A and B. Find the coordinates of point A.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

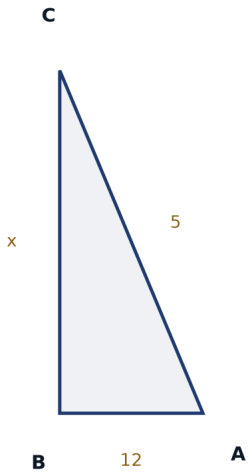
The diagram shows the graph of the quadratic function $f(x) = x^2 + ax + b$. The graph passes through the points $(-1, 8)$ and $(3, 0)$. Find the values of a and b.



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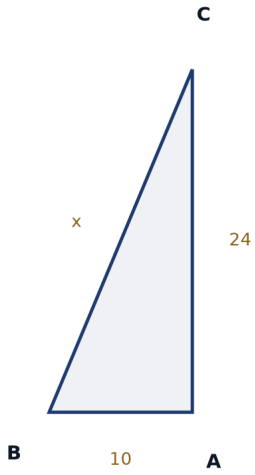
9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the right-angled triangle ABC, $\angle ABC = 90^\circ$. If $AB = 5$ cm and $BC = 12$ cm, find the length of the hypotenuse AC.



10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

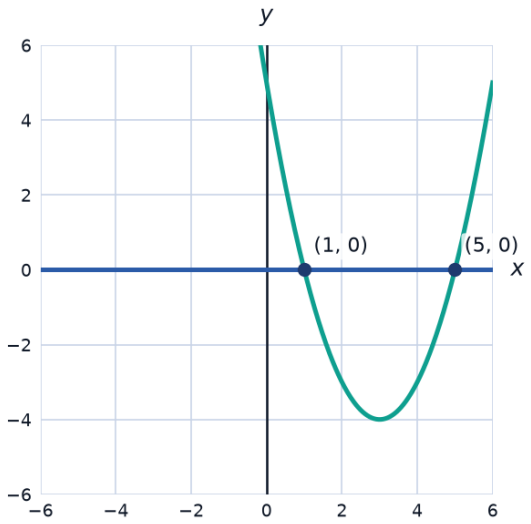
The diagram shows a right-angled triangle ABC, with the right angle at B. The length of side AB is 10 cm and the length of side BC is 24 cm. A semicircle is drawn with AC as its diameter. Calculate the area of the semicircle in terms of π.



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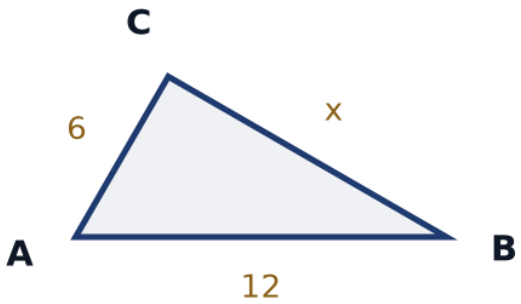
11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows the graph of the quadratic function $y = x^2 + bx + c$. The graph intersects the x-axis at $(1, 0)$ and $(5, 0)$. Find the values of b and c.



12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

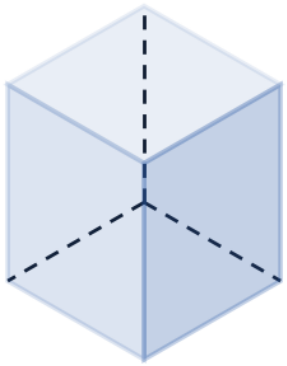
In a right-angled triangle, one of the acute angles is 30° and the side opposite to this angle is 6 cm. Calculate the length of the hypotenuse.



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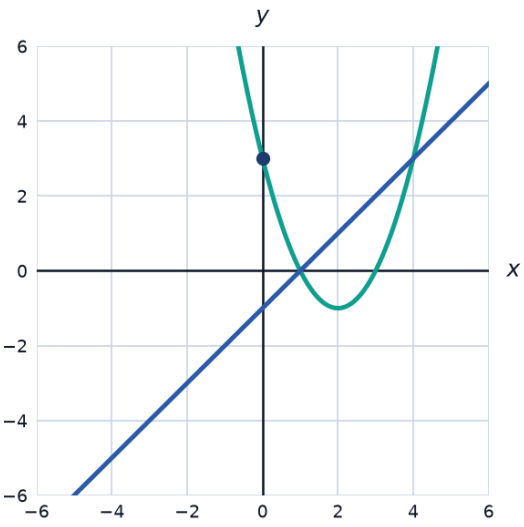
13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A rectangular prism has a length of 5 cm, a width of 3 cm, and a height of 4 cm. Calculate the volume of the prism.



14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the function $y = f(x)$ is shown below. Find the value of $f(0)$.

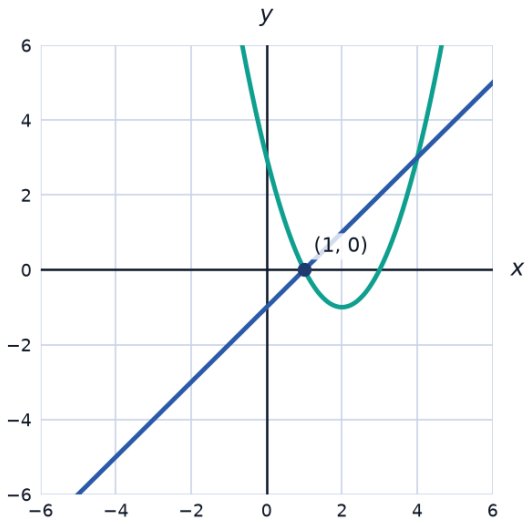


Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Trigonometry

The diagram shows the graph of the quadratic function $y = x^2 - 4x + k$. The graph intersects the x-axis at point P(1, 0) and the y-axis at point Q. Determine the value of k.



16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Transformations & Vectors

In the diagram below, lines AB and DE are parallel. Points A, C, and D are collinear, and points B, C, and E are collinear. If $AC = 6$ cm, $CD = 9$ cm, and $DE = 12$ cm, find the length of AB.

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Probability & Statistics

A point $A(3, 5)$ is mapped to $A'(1, 3)$ by a single transformation. The same transformation is then applied to a second point $B(x, y)$, mapping it to $B'(x - 2, y - 2)$. The vector representing the displacement from A to A' is $\vec{v}$. A point $C(2, 1)$ is reflected in the line $y = x$ to give point C'. Calculate the magnitude of the vector $A\vec{C}'$.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Functions

A survey of 100 students about their preferred social media platforms yielded the following information: - 55 students use Platform A. - 62 students use Platform B. - 18 students use neither Platform A nor Platform B. - x students use both Platform A and Platform B. Find the value of x.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Sets & Venn Diagrams

A triangle ABC has vertices A(2, 3), B(5, 1), and C(3, 5). This triangle is subjected to a sequence of transformations. First, it is reflected in the line $y = x$ to produce triangle A'B'C'. Then, triangle A'B'C' is translated by the vector $(-3; 1)$ to produce triangle A"B"C". ; Determine the coordinates of the vertices of triangle A"B"C".

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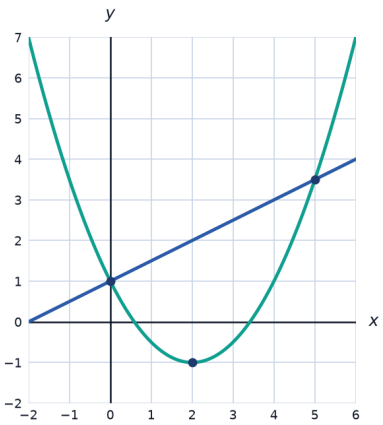
20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Algebra & Graphs

The diagram shows a point A and its image A' after a single transformation. The coordinates of A are (2, 1) and the coordinates of A' are (5, 3). The transformation can be represented by the vector $(p; q)$. Find the value of p.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 3$ in estimation, evaluate $f(2)$.



a. 7

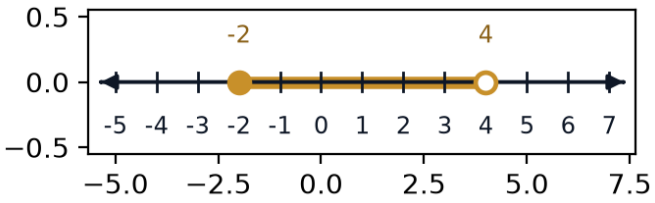
b. 9

c. 4

d. 11

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

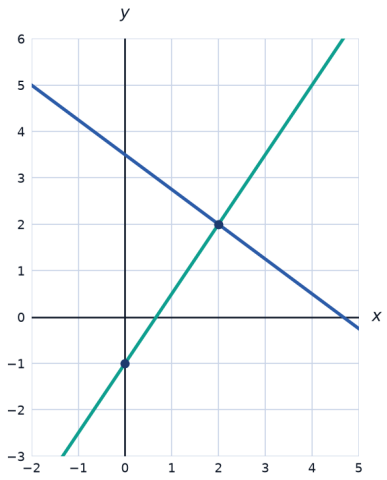
c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 4$ in rounding, evaluate $f(2)$.



a. 10

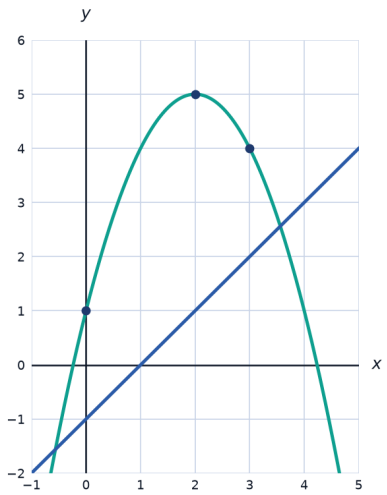
b. 13

c. 6

d. 14

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 5$ in four operations with whole numbers, evaluate $f(2)$.



a. 13

b. 17

c. 8

d. 17

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for estimation consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for rounding consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for order of operations with whole numbers consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 6$ in order of operations with whole numbers, evaluate $f(4)$.

a. 26

b. 31

c. 20

d. 30

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for using technology for operations with whole numbers consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 6x + 7$ in using technology for operations with whole numbers, evaluate $f(4)$.

a. 31

b. 37

c. 24

d. 35

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 2x + 8$ in check reasonableness of calculations, evaluate $f(3)$.

a. 14

b. 16

c. 6

d. 18

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the lowest common multiple (LCM) of 12 and 15.

a. 30

b. 45

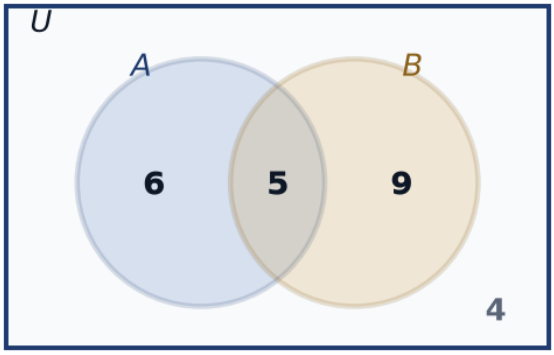
c. 60

d. 90

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Universal set $\mathscr{E} = \{x : x \text{ is an integer}, 1 \le x \le 20\}$. $P = \{x : x \text{ is a prime number}\}$ and $O = \{x : x \text{ is an odd number}\}$. What is $n(P \cap O')$?



a. 1

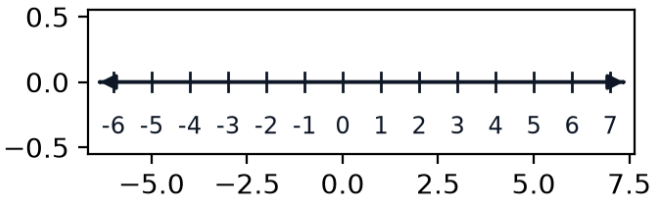
b. 2

c. 8

d. 9

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows a number line with points marked. What is the value of x?



a. -5

b. -3

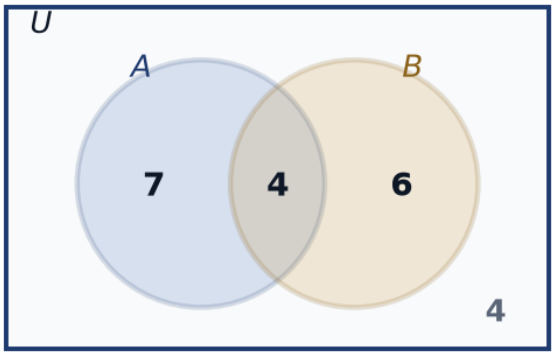
c. 1

d. 5

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Universal set $\mathscr{E} = \{x : x \in \mathbb{Z}^+, 1 \le x \le 20\}$. Set $P = \{x : x \text{ is a prime number}\}$. What is $n(P)$?



a. 8

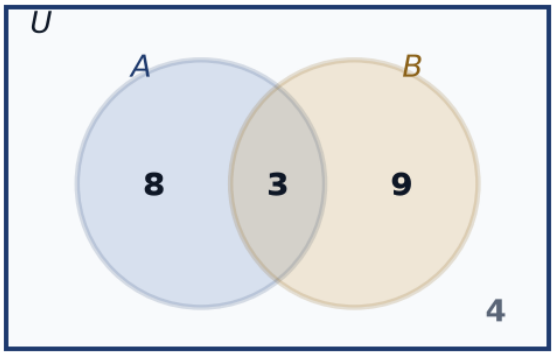
b. 7

c. 9

d. 10

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Universal set $\mathscr{E} = \{x : x \in \mathbb{Z}^+, x \le 15\}$. Set $A = \{x : x \text{ is a multiple of } 3\}$ and $B = \{x : x \text{ is a multiple of } 5\}$. Find $A \cap B$.



a. $\{15\}$

b. $\{3, 5, 15\}$

c. $\{3, 5, 6, 9, 10, 12, 15\}$

d. $\varnothing$

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In a class of 60 students, all students study at least one of the subjects French (F) or Spanish (S). The number of students who study French is $n(F)$, the number of students who study Spanish is $n(S)$, and the number of students who study both French and Spanish is $n(F \cap S)$. Given that $n(F) = 3x - 4$, $n(S) = 2x + 6$, and $n(F \cup S) = 50$, find the value of x.

a. 4

b. 5

c. 6

d. 7

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Functions

In a survey of 100 students about their preferred social media platforms, it was found that 60 use Platform A, 55 use Platform B, and 30 use both Platform A and Platform B. A further 15 students use neither Platform A nor Platform B. Let A be the set of students who use Platform A, and B be the set of students who use Platform B. (a) Represent this information on a Venn diagram. (b) Calculate the number of students who use only Platform A. (c) Calculate the number of students who use only Platform B. (d) If a student is chosen at random from the survey, what is the probability that they use Platform A or Platform B (or both)?

a. 75/100

b. 85/100

c. 95/100

d. 70/100

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Sets & Venn Diagrams

A special sequence of numbers is defined by the recurrence relation $a_{n+1} = \frac{2a_n + k}{a_n}$, where k is a positive constant. The first term is $a_1 = 3$. If $a_3 = \frac{13}{5}$, find the value of k and the value of a_4.

a. $k = 3, a_4 = 7/3$

b. $k = 4, a_4 = 13/4$

c. $k = 5, a_4 = 9/5$

d. $k = 2, a_4 = 10/3$

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Algebra & Graphs

Rationalise the denominator and simplify: $\frac{10}{3-\sqrt{5}}$.

a. $\frac{15+5\sqrt{5}}{2}$

b. $5(3+\sqrt{5})$

c. $\frac{10(3-\sqrt{5})}{4}$

d. $3+\sqrt{5}$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Coordinate Geometry

A car travels a distance of 150 km, correct to the nearest 10 km, in a time of 2 hours, correct to the nearest 0.1 hour. What is the upper bound of the average speed in km/h?

a. 79.5 km/h

b. 77.5 km/h

c. 75.0 km/h

d. 81.6 km/h

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Mensuration

A bank account pays compound interest of 3% per year. How many complete years will it take for an initial deposit of $x3000?

a. 12 years

b. 13 years

c. 14 years

d. 15 years

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Geometry

Convert $0.1\dot{4}\dot{8}$ to a fraction in its simplest form.

a. $\frac{148}{1000}$

b. $\frac{49}{330}$

c. $\frac{37}{250}$

d. $\frac{7}{45}$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

A rectangle has length $l = 8.4$ cm correct to 1 decimal place and width $w = 5.6$ cm correct to 1 decimal place. Calculate the lower bound for the area of the rectangle.

a. 47.04 cm²

b. 46.3425 cm²

c. 46.75 cm²

d. 47.7425 cm²

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Transformations & Vectors

Find the highest common factor (HCF) of 36 and 54.

a. 6

b. 9

c. 18

d. 27

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